

Team #07

ASIC - Design Implementation of Spike-NN

Mentor TA:

Akshayini K

SRN	Name
PES1UG23EC066	Aumkar Ranjan Behura
PES1UG23EC081	Chirag V Rao
PES1UG23EC117	Harini L R
PES1UG23EC106	Gagan S

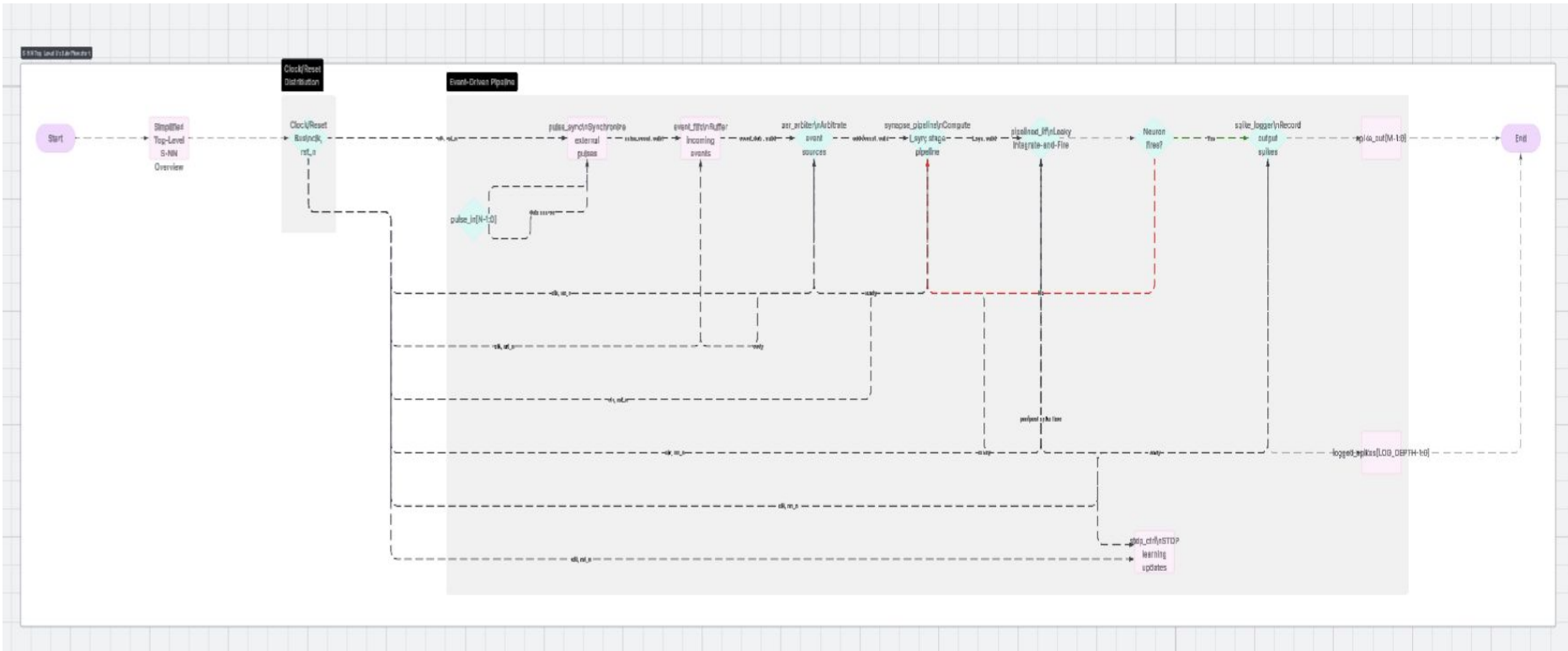
What is your topic?

- Spike-NN uses spiking neurons that fire only when membrane potential crosses a threshold.
- Implements Leaky Integrate-and-Fire (LIF) model: neurons accumulate weighted spikes, leak over time, and emit a spike on threshold crossing.
- Computation is event-driven, temporal, and sparse, unlike continuous ANN activations.

Motivation

1. Ultra-low power: SNNs compute only when spikes occur, reducing energy and memory accesses—perfect for battery-powered edge devices.
2. Low latency: Event-driven processing lets spikes propagate instantly, enabling real-time response unlike frame-based ANNs.
3. Neuromorphic scalability: Simple neuron/synapse circuits tile efficiently on ASICs, enabling compact, parallel, brain-like hardware.

Architecture Block Diagram

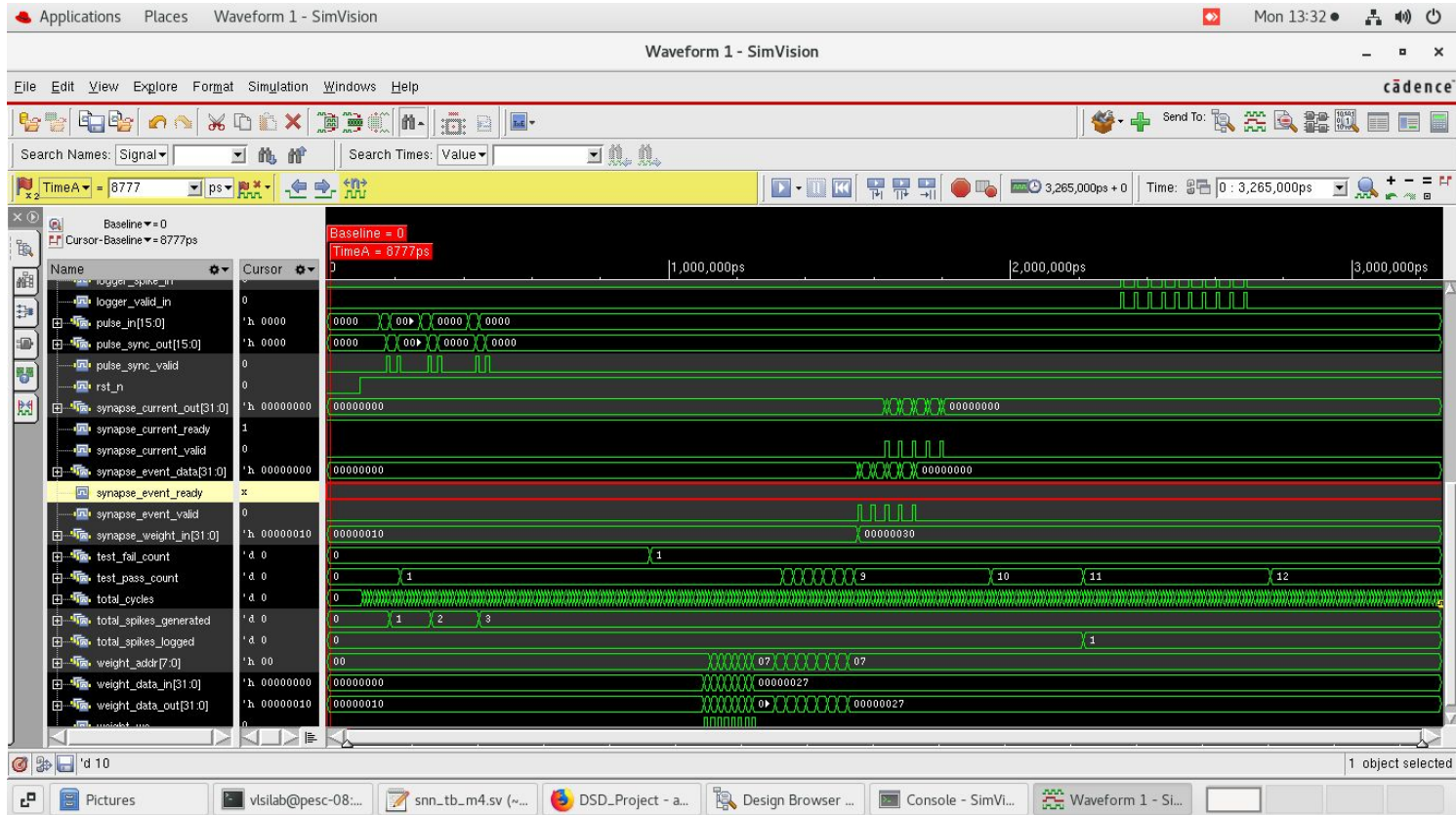


Contribution

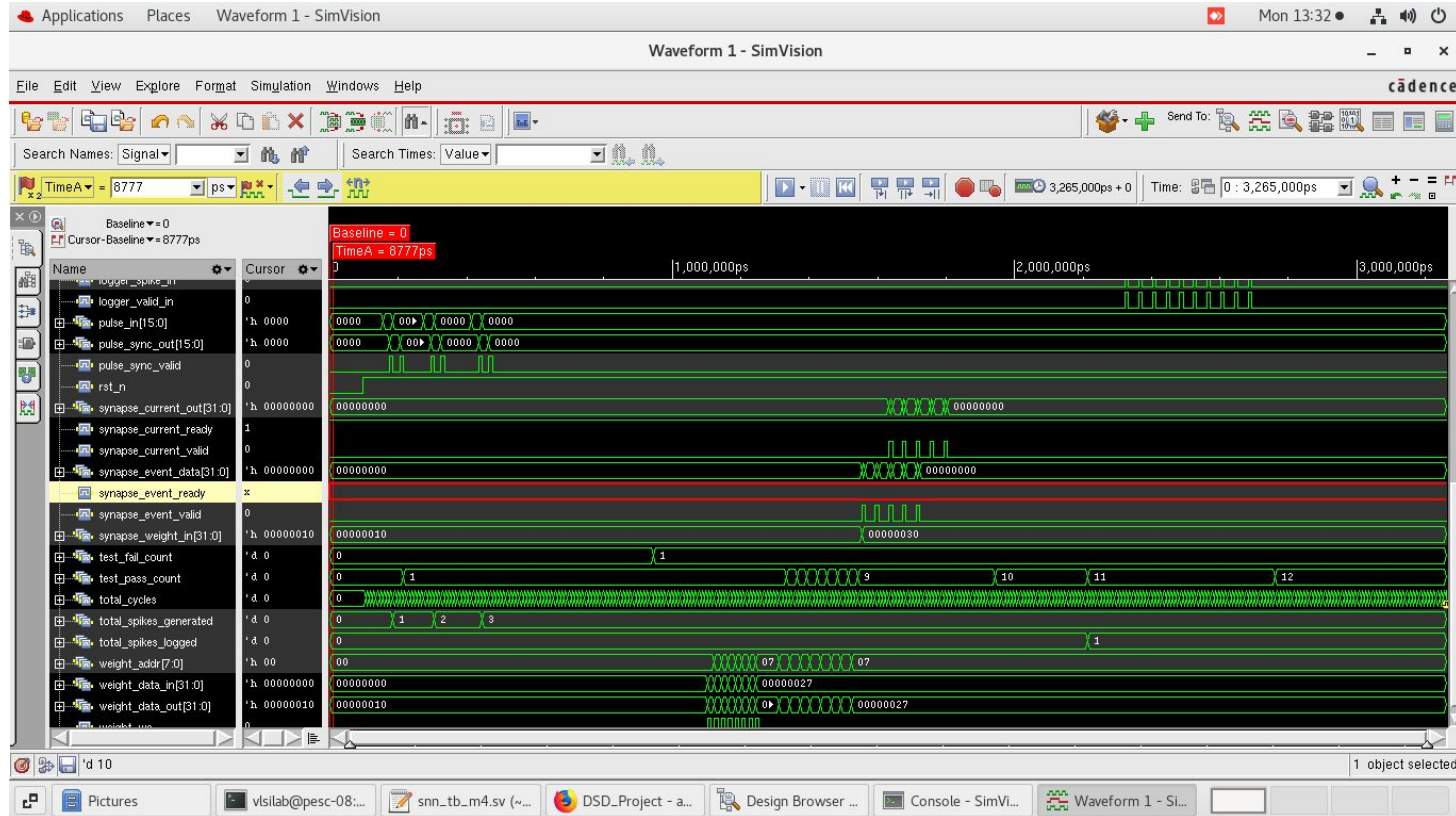
Individual Contributions-

Aumkar	pipelined_lif + event_fifo
Harini	weight_ram_par + synapse_pipeline
Chirag	Pulse_sync + weight_ram_par
Gagan	Spike_logger + Testbench

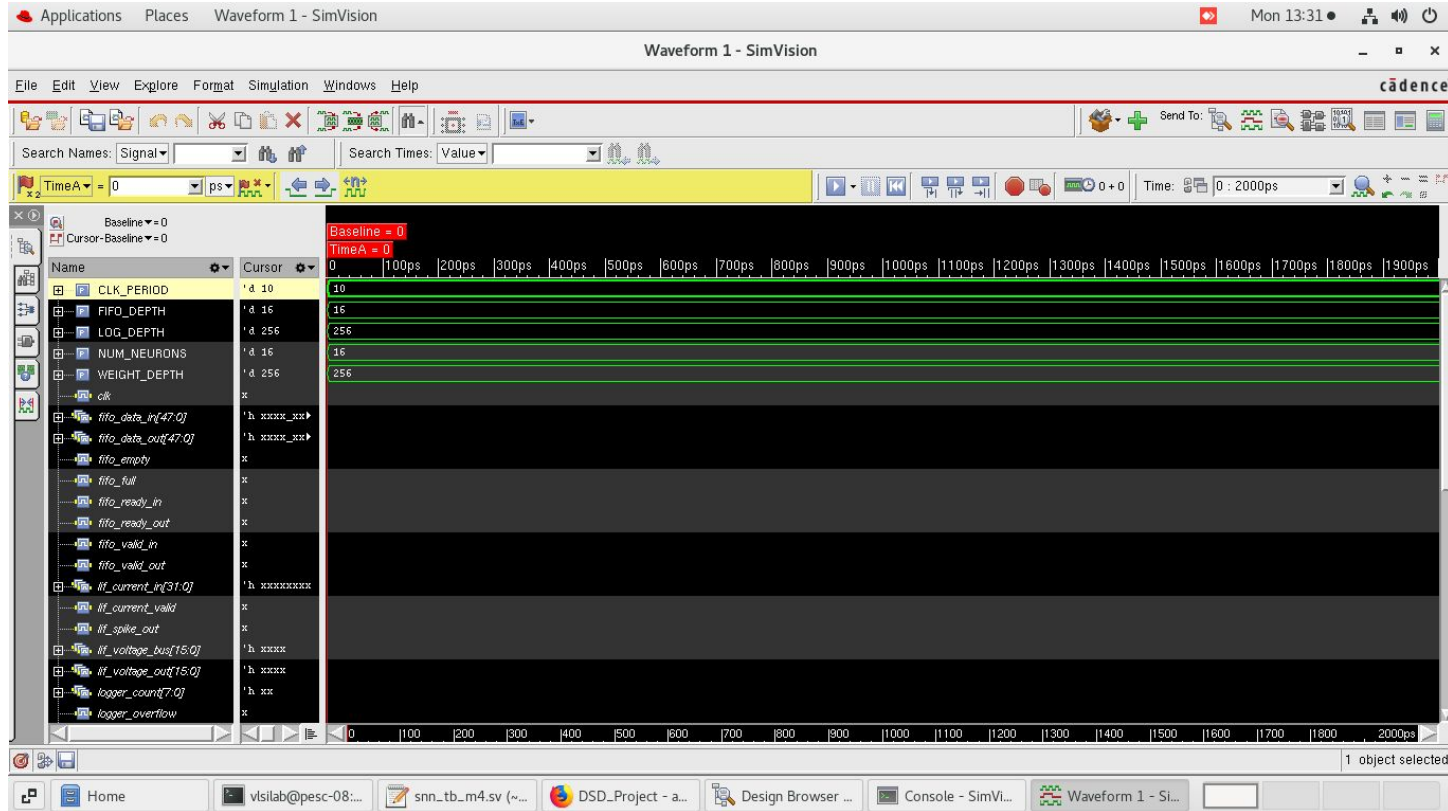
Waveforms (and explanation of design)



Waveforms (and explanation of design)



Waveforms (and explanation of design)



Area Report

```
@genus:design:lif_system_wrapper 12> report_area
r incomplete. [RPT-80]
```

Wai

: The design design:lif_system_wrapper should be mapped to get accurate area details.

```
=====
Generated by:      Genus(TM) Synthesis Solution 20.11-s111_1
Generated on:      Nov 18 2025  02:14:11 pm
Module:           lif_system_wrapper
Technology library: saed90nm_typ updated 30.Sep.2011
Operating conditions: _nominal_ (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
lif_system_wrapper		8	162.202	0.000	162.202	<none> (D)
mux_73_30	bmux	8	162.202	0.000	162.202	<none> (D)

(D) = wireload is default in technology library

Power Report

Instance: /lif_system_wrapper

Power Unit: W

PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.45259e-07	7.56506e-06	0.00000e+00	8.11031e-06	100.00%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	5.45259e-07	7.56506e-06	0.00000e+00	8.11031e-06	100.00%
Percentage	6.72%	93.28%	0.00%	100.00%	100.00%

Power Report

Instance: /lif_system_wrapper

Power Unit: W

PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	5.45259e-07	3.95317e-06	0.00000e+00	4.49843e-06	100.00%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	5.45259e-07	3.95317e-06	0.00000e+00	4.49843e-06	100.00%
Percentage	12.12%	87.88%	0.00%	100.00%	100.00%

Timing Report

Max Freq of your design:

Screenshot of Report at Max Freq
No Text

References

- <https://cepdnack.github.io/e18-4yp-Neuromorphic-NoC-Architecture-for-SNNs/>
- <https://open-neuromorphic.org/blog/spiking-neurons-digital-hardware-implementation/>
- <https://github.com/amckenna41/izhikevich-spiking-neuron>
- https://github.com/DeltaProxima/Spiking_NN_Verilog
- <https://github.com/allenblade-commits/SpikingNeuralNetworkusingLIFNeuron>
- <https://github.com/nikitabuzov/SpikingNeuralNet>
- [Reference Papers](#)